

# Non-Uniform Reanchoring

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# Online vs Offline Reanchoring

- Online reanchoring: anchor positions are selected dynamically during generation
  - Main issue: Decisions are made without knowledge of future timesteps, so an anchor chosen now may not be optimal later.
- Offline reanchoring: anchor positions are fixed using a calibration dataset and reused for all prompts.
  - Main issue: may not generalize well to different prompts

# Online Reanchoring Algorithm

- For every submodule per layer calculate the relative quantization error (RQE):

$$\text{Relative Quantization Error}(t) = \frac{\|\Delta_t - \hat{\Delta}_t\|_F^2}{\|X_t\|_F^2}$$

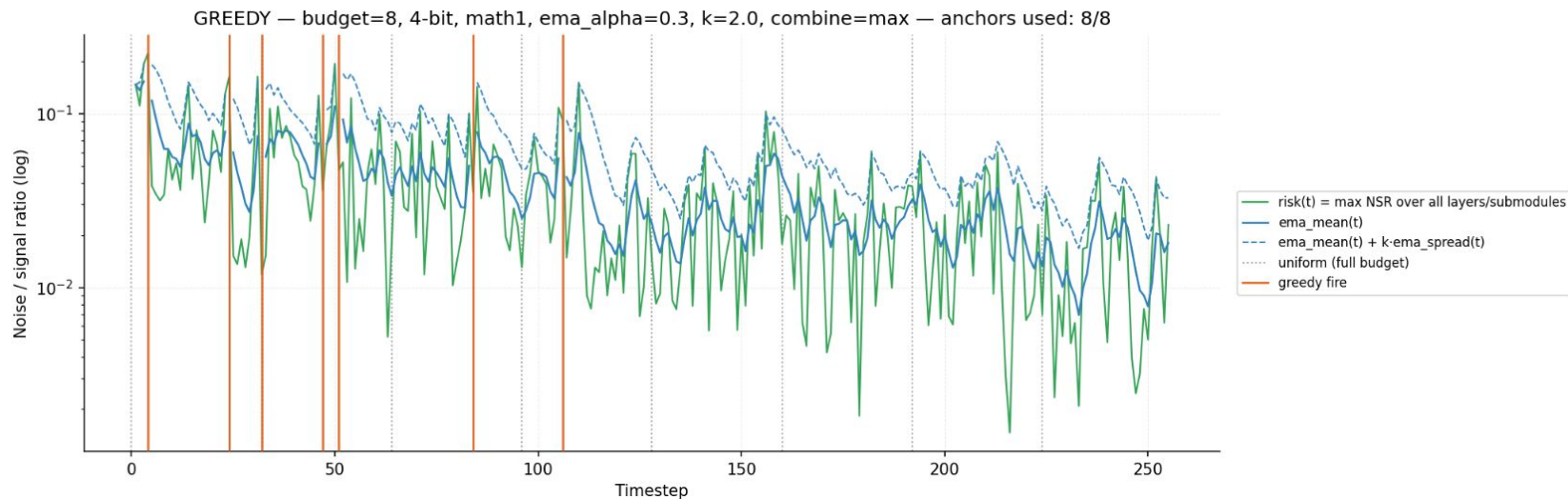
- Find the max RQE across all layers
- Update a running estimated moving average mean and spread using these equations:

$$\text{EMA}_{\text{mean}}(t) = \alpha \cdot \text{RQE}(t) + (1 - \alpha) \cdot \text{EMA}_{\text{mean}}(t - 1)$$

$$\text{EMA}_{\text{spread}}(t) = \alpha \cdot |\text{RQE}(t) - \text{EMA}_{\text{mean}}(t)| + (1 - \alpha) \cdot \text{EMA}_{\text{spread}}(t - 1)$$

- Add an anchor if it exceeds the threshold:  $\text{Threshold}(t) = \text{EMA}_{\text{mean}}(t) + k \cdot \text{EMA}_{\text{spread}}(t)$

# Online Reanchoring Example



# Online Reanchoring Evaluation - QuaRot

LLaDA-8B-Instruct accuracy under W4A4 delta-quant on top of QuaRot rotations: rotation-only baseline vs. full-precision anchors spread uniformly vs. non-uniformly placed by online anchor policy

| Anchors | Placement   | Math  |         |       |                      | Code      |       |       |                      | Overall | $\Delta\%$           |
|---------|-------------|-------|---------|-------|----------------------|-----------|-------|-------|----------------------|---------|----------------------|
|         |             | GSM8K | MATH500 | Avg   | $\Delta\%$           | HumanEval | MBPP  | Avg   | $\Delta\%$           |         |                      |
| –       | FP16        | 78.85 | 34.87   | 56.86 | –                    | 39.63     | 39.80 | 39.72 | –                    | 48.29   | –                    |
| –       | QuaRot      | 67.55 | 25.34   | 46.45 | $\downarrow 18.32\%$ | 26.83     | 32.00 | 29.41 | $\downarrow 25.94\%$ | 37.93   | $\downarrow 21.45\%$ |
| 2       | Uniform     | 58.45 | 18.51   | 38.48 | $\downarrow 32.33\%$ | 22.56     | 31.20 | 26.88 | $\downarrow 32.33\%$ | 32.68   | $\downarrow 32.33\%$ |
|         | Non-uniform | 53.83 | 17.93   | 35.88 | $\downarrow 36.90\%$ | 26.22     | 33.40 | 29.81 | $\downarrow 24.95\%$ | 32.84   | $\downarrow 31.98\%$ |
| 4       | Uniform     | 64.14 | 21.49   | 42.81 | $\downarrow 24.70\%$ | 23.17     | 31.20 | 27.19 | $\downarrow 31.56\%$ | 35.00   | $\downarrow 27.52\%$ |
|         | Non-uniform | 63.08 | 21.23   | 42.16 | $\downarrow 25.86\%$ | 28.05     | 34.40 | 31.23 | $\downarrow 21.39\%$ | 36.69   | $\downarrow 24.02\%$ |
| 8       | Uniform     | 68.46 | 24.99   | 46.72 | $\downarrow 17.82\%$ | 29.27     | 33.80 | 31.53 | $\downarrow 20.61\%$ | 39.13   | $\downarrow 18.97\%$ |
|         | Non-uniform | 68.69 | 24.15   | 46.42 | $\downarrow 18.36\%$ | 28.66     | 34.80 | 31.73 | $\downarrow 20.12\%$ | 39.08   | $\downarrow 19.08\%$ |
| 16      | Uniform     | 69.75 | 26.76   | 48.26 | $\downarrow 15.13\%$ | 31.71     | 34.20 | 32.95 | $\downarrow 17.03\%$ | 40.61   | $\downarrow 15.91\%$ |
|         | Non-uniform | 69.98 | 25.40   | 47.69 | $\downarrow 16.13\%$ | 29.27     | 34.60 | 31.94 | $\downarrow 19.60\%$ | 39.81   | $\downarrow 17.56\%$ |
| 32      | Uniform     | 72.18 | 27.54   | 49.86 | $\downarrow 12.31\%$ | 34.76     | 35.20 | 34.98 | $\downarrow 11.93\%$ | 42.42   | $\downarrow 12.16\%$ |
|         | Non-uniform | 70.28 | 25.40   | 47.84 | $\downarrow 15.86\%$ | 29.27     | 34.60 | 31.94 | $\downarrow 19.60\%$ | 39.89   | $\downarrow 17.40\%$ |

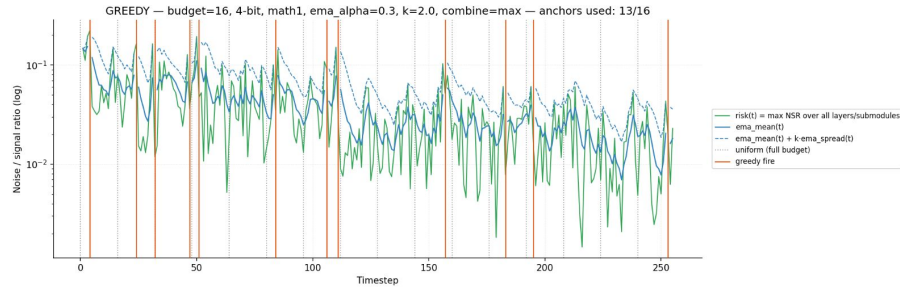
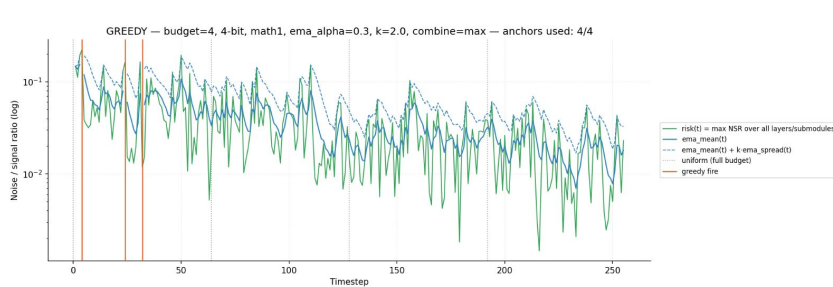
# Online Reanchoring Evaluation - DuQuant

LLaDA-8B-Instruct accuracy under W4A4 delta-quant on top of DuQuant rotations: rotation-only baseline vs. full-precision anchors spread uniformly vs. non-uniformly placed by online anchor policy

| Anchors | Placement   | Math  |         |       |                      | Code      |       |       |                      | Overall | $\Delta\%$           |
|---------|-------------|-------|---------|-------|----------------------|-----------|-------|-------|----------------------|---------|----------------------|
|         |             | GSM8K | MATH500 | Avg   | $\Delta\%$           | HumanEval | MBPP  | Avg   | $\Delta\%$           |         |                      |
| –       | FP16        | 78.85 | 34.87   | 56.86 | –                    | 39.63     | 39.80 | 39.72 | –                    | 48.29   | –                    |
| –       | DuQuant     | 75.74 | 32.75   | 54.24 | $\downarrow 4.60\%$  | 35.37     | 29.80 | 32.59 | $\downarrow 17.96\%$ | 43.41   | $\downarrow 10.10\%$ |
| 2       | Uniform     | 61.11 | 20.62   | 40.87 | $\downarrow 28.13\%$ | 20.12     | 24.40 | 22.26 | $\downarrow 43.96\%$ | 31.56   | $\downarrow 34.64\%$ |
|         | Non-uniform | 52.99 | 19.23   | 36.11 | $\downarrow 36.49\%$ | 23.78     | 28.80 | 26.29 | $\downarrow 33.81\%$ | 31.20   | $\downarrow 35.39\%$ |
| 4       | Uniform     | 69.22 | 25.49   | 47.35 | $\downarrow 16.72\%$ | 23.78     | 24.60 | 24.19 | $\downarrow 39.10\%$ | 35.77   | $\downarrow 25.92\%$ |
|         | Non-uniform | 62.09 | 22.33   | 42.21 | $\downarrow 25.77\%$ | 31.10     | 30.80 | 30.95 | $\downarrow 22.08\%$ | 36.58   | $\downarrow 24.25\%$ |
| 8       | Uniform     | 73.39 | 29.74   | 51.56 | $\downarrow 9.31\%$  | 32.32     | 27.40 | 29.86 | $\downarrow 24.82\%$ | 40.71   | $\downarrow 15.69\%$ |
|         | Non-uniform | 71.65 | 27.63   | 49.64 | $\downarrow 12.70\%$ | 35.98     | 30.80 | 33.39 | $\downarrow 15.94\%$ | 41.52   | $\downarrow 14.03\%$ |
| 16      | Uniform     | 73.31 | 32.03   | 52.67 | $\downarrow 7.37\%$  | 30.49     | 29.00 | 29.74 | $\downarrow 25.11\%$ | 41.21   | $\downarrow 14.67\%$ |
|         | Non-uniform | 73.69 | 29.51   | 51.60 | $\downarrow 9.25\%$  | 37.20     | 31.20 | 34.20 | $\downarrow 13.90\%$ | 42.90   | $\downarrow 11.16\%$ |
| 32      | Uniform     | 75.21 | 32.96   | 54.08 | $\downarrow 4.88\%$  | 37.80     | 33.00 | 35.40 | $\downarrow 10.88\%$ | 44.74   | $\downarrow 7.35\%$  |
|         | Non-uniform | 73.69 | 29.63   | 51.66 | $\downarrow 9.15\%$  | 37.20     | 31.00 | 34.10 | $\downarrow 14.15\%$ | 42.88   | $\downarrow 11.20\%$ |

# Discussion of Online Reanchoring Results

- Results from previous plots show that this online reanchoring approach does not beat the uniform method
- It is failing due to two main reasons:
  - Most of the anchors are being used at earlier timesteps leaving no budget for later timesteps
  - The  $k$  anchors may not be fully utilized if the threshold is not exceeded  $k$  times

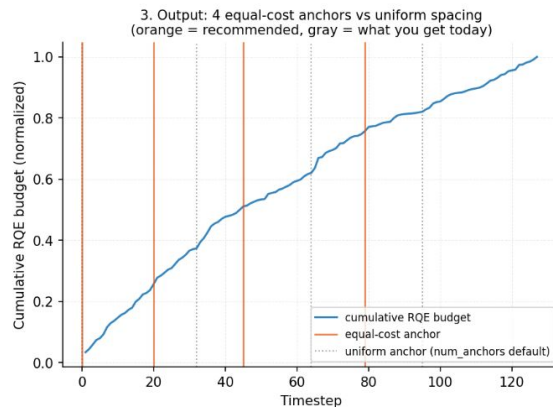
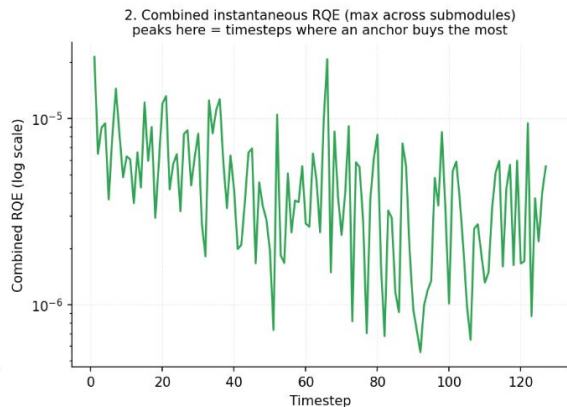
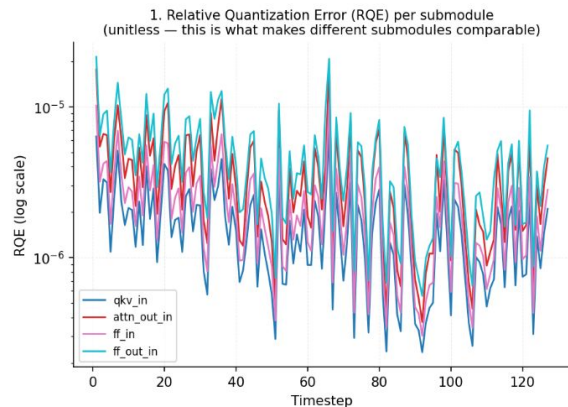


# Offline Reanchoring Algorithm

- Run the LLaDa-8B-instruct model on 16 calibration samples from wikitext2 dataset
- At every timestep and for every module calculate the relative quantization error (RQE)
- Average the RQE over layers for each timestep and submodule and over all calibration prompts
- Choose the max RQE over submodules for each timestep
- Calculate the cumulative RQE budget over time
- Divide this cumulative RQE budget into K segments of equal costs and place the K anchors at the boundaries of the segments

# Offline Reanchoring Example

Reanchor schedule derivation, 8-bit, budget=4, combine=max



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| 2       | Uniform     | 58.45 | 18.51   | 38.48 | ↓ 32.33%   | 22.56     | 31.20 | 26.88 | ↓ 32.33%   | 32.68   | ↓ 32.33%   |
|         | Non-uniform | 59.36 | 20.39   | 39.88 | ↓ 29.87%   | 23.17     | 31.00 | 27.09 | ↓ 31.80%   | 33.48   | ↓ 30.67%   |
| 4       | Uniform     | 64.14 | 21.49   | 42.81 | ↓ 24.70%   | 23.17     | 31.20 | 27.19 | ↓ 31.56%   | 35.00   | ↓ 27.52%   |
|         | Non-uniform | 66.34 | 23.14   | 44.74 | ↓ 21.32%   | 28.66     | 32.20 | 30.43 | ↓ 23.38%   | 37.59   | ↓ 22.17%   |
| 8       | Uniform     | 68.46 | 24.99   | 46.73 | ↓ 17.82%   | 29.27     | 33.80 | 31.53 | ↓ 20.60%   | 39.13   | ↓ 18.97%   |
|         | Non-uniform | 68.69 | 25.69   | 47.19 | ↓ 17.01%   | 33.54     | 33.40 | 33.47 | ↓ 15.73%   | 40.33   | ↓ 16.48%   |
| 16      | Uniform     | 69.75 | 26.76   | 48.25 | ↓ 15.13%   | 31.71     | 34.20 | 32.95 | ↓ 17.03%   | 40.61   | ↓ 15.91%   |
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|         | Non-uniform | 71.04 | 28.53   | 49.79 | ↓ 12.44%   | 35.98     | 37.80 | 36.89 | ↓ 7.12%    | 43.34   | ↓ 10.26%   |

# Offline Reanchoring Evaluation - DuQuant

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|         | Non-uniform | 73.46 | 30.90   | 52.18 | $\downarrow 8.23\%$  | 31.10     | 29.20 | 30.15 | $\downarrow 24.09\%$ | 41.16   | $\downarrow 14.75\%$ |
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|         | Non-uniform | 75.28 | 32.78   | 54.03 | $\downarrow 4.98\%$  | 36.59     | 33.00 | 34.79 | $\downarrow 12.40\%$ | 44.41   | $\downarrow 8.03\%$  |

# Discussion of Offline Reanchoring Results

- For both QuaRot and DuQuant the offline non-uniform method beats the uniform method on coding benchmarks for the 16 anchor count
- QuaRot non-uniform also beats the uniform method for the coding tasks on 32 anchors
- For 32 anchors, 256 timesteps, there is 1 anchor every 8 steps so the compute speedup is  $8 \cdot T / (7 \cdot (T/2) + T) = 1.78x$  compared to FP16
- For 16 anchors, 256 timesteps there is 1 anchor every 16 steps so compute speedup is  $16 \cdot T / (15 \cdot (T/2) + T) = 1.88x$  compared to FP16
- Offline reanchoring for DuQuant can achieve uniform results for 32 anchors using only 16 anchors this leads to 5.62 % increase in compute speed at the cost of 1.53% reduction in average code benchmark accuracy

# Plan For Next Week

- Find better metrics for both offline and online reanchoring approaches and conduct experiments based on them
- For online methods deal with issues related to either under-utilization of the anchor budget or fully using it early timesteps.
- The goal is to further improve the accuracy of high anchor W4A4 non-uniform methods